TEACHER GUIDE · GRADE 1 · UNPLUGGED SEQUENCING

What Must Come First?

Quick facilitation notes & answer key

Goal

Students reason about dependencies in an algorithm: which step must come before another, which step cannot go early, and which steps are flexible — then write a working order. Describing how the order of steps affects whether the algorithm works is C3.2; writing a valid order is C3.1.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: act out the tea steps so the dependencies are felt (you cannot pour water that is not boiled yet).

Run it (about 20 min)

1. I DO: 'To pour the water it has to be hot — so boiling the water must come before pouring.' That is a dependency.

2. WE DO: Exercise 1 — boil the water (B) must come before pour (D). Exercise 2 — you cannot add milk (A) before pouring (D); there is nothing in the cup yet.

3. YOU DO: Exercise 3 — boil the water (B) and put the tea bag in the cup (C) can go in either order; a working order is B, C, D, A.

4. Connect: 'Coders look for what must come first so the steps don't break each other.'

Answer key (modelled by the run-gate)

Ex 1 — B (boil the water) must come before D (pour the water).

Ex 2 — NO: you cannot add the milk before pouring the water, because there is no tea in the cup to add milk to yet. A must come after D.

Ex 3 — B and C can be done in either order (neither depends on the other). A working order: B, C, D, A (or C, B, D, A).

Watch for & differentiate

Common idea: students may think there is only one right order — point out the flexible pair (B and C) to show some steps don't depend on each other.

Simplify: focus on Exercise 1 and 2 only (one dependency at a time).

Extend: ask which step must always be LAST (add the milk) and why.

Success indicator: the child names B as must-come-first, explains why milk can't go early, finds the B/C flexible pair, and writes a working order.